

# Glossary

## ACID

A substance with a pH lower than 7.

## ALGAE

Plantlike organisms that can make food using energy from sunlight.

## ALKALI

See Base

## ALLOY

A mixture of two or more metals, or of a metal and a non-metal.

## ANALOG

Relating to signals or information represented by a continuously varying value, such as a wave.

## ATMOSPHERE

The layer of breathable gases, such as oxygen and nitrogen, that surrounds Earth.

## ATOM

The smallest unit of an element.

## BACTERIA

Microscopic organisms with a simple, single-celled form.

## BASE

A substance with pH higher than 7. Bases that are soluble in water are called alkalis. Also: one of the four chemicals that make up the "rungs" of a DNA double helix.

## BIOLOGY

The science of living things.

## BUOYANCY

The tendency of a solid to float or sink in liquids.

## CARBOHYDRATE

An energy-rich substance, such as sugar or starch.

## CATALYST

A substance that makes a chemical reaction occur much more rapidly, but is not changed by the reaction.

## CELL

The smallest unit of life.

## CHEMICAL BOND

An attraction between particles, such as atoms or ions.

## CHEMICAL REACTION

A process that changes substances into new substances by breaking and making chemical bonds.

## CHEMISTRY

The science of matter and elements.

## CHROMOSOME

A threadlike structure, found in the nucleus of cells, that is made up of coiled strands of DNA. Humans have 46 chromosomes per body cell.

## CLIMATE

The most common weather conditions in an area over a long period of time.

## COMBUSTION

A chemical reaction in which a substance reacts with oxygen, releasing heat and flames.

## COMPOUND

A chemical substance in which two or more elements have bonded together.

## CONCENTRATION

The amount of one substance mixed in a known volume of the other.

## CONDENSATION

A process whereby a gas changes into a liquid.

## CONDUCTOR

A substance through which heat or electric current flows easily.

## COVALENT BOND

A type of chemical bond in a molecule where atoms share one or more electrons.

## DNA

A material found in the cells of all organisms that carries instructions for how a living thing will look and function.

## DRAG

The resistance force formed when an object pushes through a fluid, such as air or water.

## ECOSYSTEM

A community of organisms and the nonliving environment around them.

## ELECTRIC CHARGE

How positive or negative a particle is.

## ELECTRON

One of the tiny particles inside an atom. It has a negative electric charge.

## ELEMENT

A simple substance made of atoms that are all the same kind.

## ENERGY

What enables work to be done. Energy exists in many different forms and cannot be created or destroyed, only transferred.

## ENZYME

A substance produced in living organisms that acts as a catalyst and speeds up chemical reactions.

## EROSION

A process by which Earth's surface rocks and soil are worn away by wind, water, or ice.

## EVAPORATION

A process by which a liquid changes into a gas.

## EVOLUTION

The process by which Earth's species gradually change over long periods of time, such as millions of years, to produce new species.

## EXCRETION

The process by which living organisms expel or get rid of waste produced by cells of the body.

## FERTILIZATION

The joining of male and female sex cells so they develop into new life.

## FISSION

A splitting apart; nuclear fission is the splitting of the nucleus of an atom.

## FOSSIL

The preserved remains or impressions of life from an earlier time.

## FOSSIL FUEL

A substance formed from the remains of ancient organisms that burns easily to release energy.

## FRICTION

The dragging force that occurs when one object moves over another.

## FUSION

A joining together; nuclear fusion is the joining of two atomic nuclei.

## GAS

A state of matter that flows to fill a container, and can be compressed.

## GENE

One of the tiny units carried on DNA that determine what a living thing looks like and how it functions.

## GLUCOSE

A simple carbohydrate, or sugar, made by photosynthesis and then used by cells as a source of energy.

## GRAVITY

The force that attracts one object to another and prevents things on Earth from floating off into space.

## HABITAT

The area where an animal naturally makes its home.

## INHERITANCE

The range of natural characteristics passed on to offspring by parents.

## INSULATOR

A material that stops heat moving from a warm object to a colder one.

